chemically identical with the parent substance. In such a case, it would be logical to expect a separation to be impossible. Nevertheless, it is feasible under certain circumstances; namely, when it is simply a case of the absorption of a neutron, the excitation energy is emitted in the form of a photon, as gamma radiation. But the latter produces a recoil in the nucleus which may cause the atom subsequently to become electrically charged, or to be torn out of its chemical bond. In this case, skilful chemical operations will effect the separation of the radioactive atoms from the other atoms. Such methods have been developed by *Szilard* and *Chalmers*, and others.

V. ARTIFICIAL RADIOACTIVE SUBSTANCES AS TRACERS

We have mentioned a few ways of utilizing artificial radioactive substances. But there is still another application of these substances which has been widely utilized during recent years, and which may be said to be the most important one at the present time. This application consists in the use of radioactive atoms as *tracers.* This is what is meant: Formerly, the identification, at some later instant, of individual atoms of a particular element was quite impossible. The reason for this was that the path followed by these atoms in biological or chemical processes could not be followed in detail because of the presence of other atoms of the same kind in the substance or organism being studied. Now, however, it is possible to attach a label to any element—as a ring is fastened to a leg of a carrier pigeon or migratory bird. This label is radioactivity which enables the path of the element to be followed in all its details.

We may illustrate this method by a simple example: Let us assume that we wish to study the diffusion of the atoms of a solid substance within the substance itself—for example, the diffusion of lead atoms in lead. Prior to the discovery of radioactivity, this would have been impossible, for one would never have been able to recognize an individual atom or to distinguish it from other lead atoms. But today, if a piece of lead containing radioactive atoms is brought into close contact with another piece which does not contain any, the lead atoms will be reciprocally interchanged between the two pieces by diffusion, and gradually increasing numbers of radioactive atoms will be discovered in places where originally there was no radioactivity at all. In this way we can obtain information concerning the velocity with which lead atoms diffuse in solid lead.

Let us take another example, the enormous practical value of which will perhaps be even more illuminating: In testing the filter of a gas mask, the main object is to determine to what degree it absorbs the poisonous substances, against which it is intended to serve as a protection. This can be